温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Pandas I

Dawei Chen

Week 3-1, 16 Sep 2025

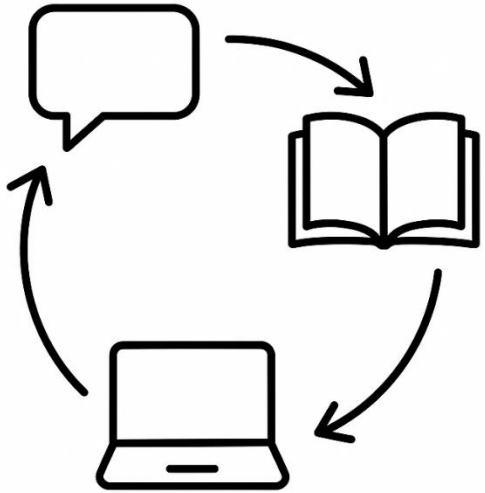
Lessons from Last Mini Task

- Knowing what code does is different from writing and running your own
- Simple steps (e.g., printing a variable, running code cells in Jupyter Notebook) can still feel tricky
- Practice and Q&A show us what we understand, and what’s still unclear
- The hardest part, getting started, is already behind you

Generated by ChatGPT5

Moving Forward

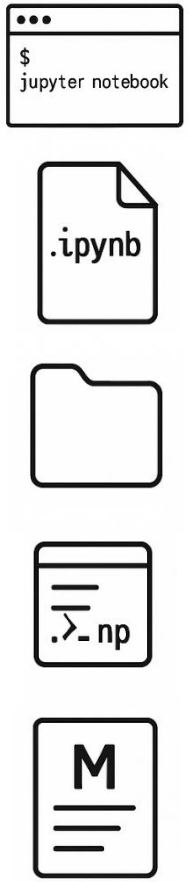
- More hands-on labs: I'll show, then you try
- Mini tasks at the end of class to build confidence step by step
- New concepts explained when we actually use them
- Explore tutorials or videos outside class at your own pace
- Practice, Q&A and feedback guide our journey



Generated by ChatGPT5

Steps to Use Jupyter Notebook

1. Open Jupyter Notebook
   - run jupyter notebook in terminal
   - open from Anaconda Navigator
2. Navigate to your working folder
3. Create a new .ipynb file
4. Rename and save the file in your folder
5. Set up the notebook
   - import modules (e.g., import pandas as pd, from math import pi)
   - define paths, random seed, etc.
6. Use Markdown
   - add heading, notes, and explanations to organize your notebook



Generated by ChatGPT5

Working with Jupyter Notebook

- Variables and Output
  - variables store values for later use
  - use print(variable) or just variable in Jupyter Notebook to display results
- Comments
  - Inline: # your note here
  - Whole cell: switch cell type to Markdown
- Code Cell Numbers
  - In [1]: first executed cell in current session
  - In [*]: cell is running
  - Markdown cells have no number
- Running Cells
  - Select a cell → click “Run” or press Shift+Enter, only the selected cell is executed
  - After restarting kernel: (1) numbers stay as before, (2) new runs start again from In [1]:

Pandas – Essential Functionality

Label → 0 1 2 3

Column

		A	B	C	D
0	a	1	2	3	4
1	b	5	6	7	8
2	c	9	10	11	12
3	d	13	14	15	16

Index

Many functions can achieve the same result. You don't need to remember them all—learn one method that works and use it well.

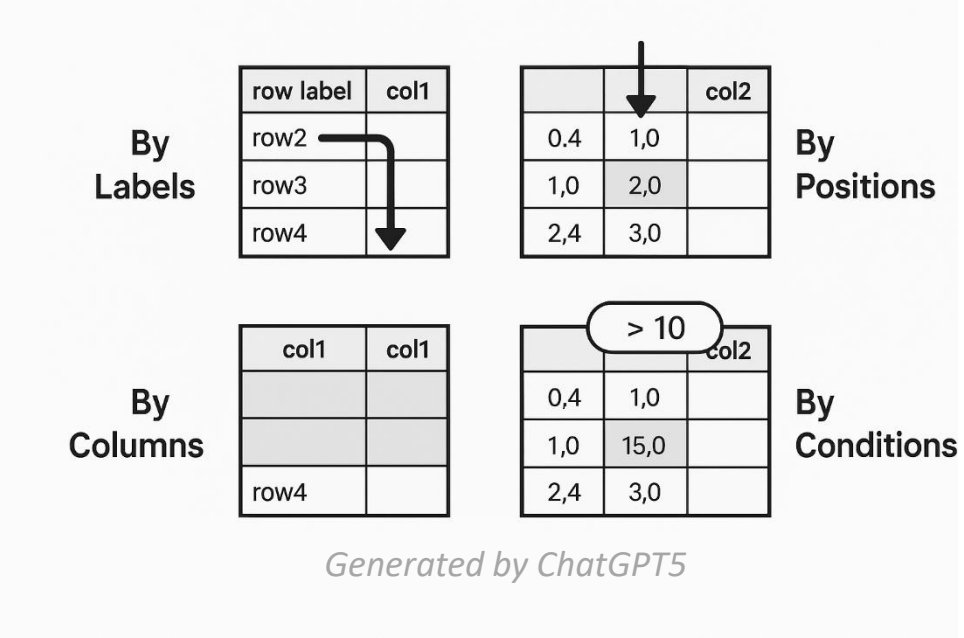
- Reindexing & Dropping
  - Rename, reorder, add, drop, sort, rank rows or columns
- Selecting & Filtering
  - Select and extract subsets based on conditions (e.g., values in a column)
- Arithmetic & Alignment
  - Arithmetic between Series or DataFrames
  - Automatic alignment on row & column labels
- Function Application & Mapping
  - Apply functions elementwise or along rows/columns or mapping function
  - Aggregate and summarize data
- Combination
  - Merge, join, concatenate two DataFrames vertically or horizontally

Pandas – Reindexing

- Rename & Reorder
  - *df.rename(columns={…}, index={…})* rename rows/columns
- Add Columns
  - *df[“new_col”] = values*
- Drop Rows/Columns
  - *df.drop[“col_name”, axis=1]* drop a column
  - *df.drop([“row1”, “row2”], axis=0)* drop rows
- Note
  - “new_col” or ‘new_col ’ → string (字符串)
  - new_col → variable (变量)

Pandas – Selecting and Filtering

- By Labels
  - *df.loc["row_label", "col_label"]* select rows/columns by names
- By Positions
  - *df.iloc[2, 0]* select rows/columns by integer index
- By Columns
  - Single column: *df["col"]*
  - Multiple columns: *df[["col1", "col2"]]*
- By Conditions
  - Boolean filtering: *df[df["col"] > 10]*
  - Combine conditions with *&* (and), *|* (or)
- By Slicing
  - Row slices: *df[2:5]*
  - With labels: *df.loc["row1":"row5"]*



Pandas – Arithmetic & Alignment

- Elementwise Operations
  - Add, subtract, multiply, divide directly on columns/rows
  - Example: *df["col1"] + df["col2"]*
- Automatic Alignment
  - Operations align on row/column labels
  - Missing values → result is *NaN*
- With Scalars
  - Apply arithmetic to all values
  - Example: *df["col1"] * 100*
- Function Application
  - Apply NumPy functions: *np.sqrt(df)*
  - Apply DataFrame methods: *df.mean(), df.sum()*
- Creating New Variables
  - Combine columns to form new variables
  - Example: *df["total"] = df["sales"] * df["price"]*

Demonstration

Don't just memorize – get familiar through practice.

Mini Task – Practicing DataFrame Basics

- Create a DataFrame with the Following Columns
  - “name”: Zhangsan, Lisi, Wangwu, Zhaoliu
  - “age”: 20, 19, 21, 20
  - “score”: 78, 89, 55, 61
- Try the Following Operations
  - Rename the column “score” to “final_score”
  - Drop the “age” column
  - Display only “name” column
  - Display “name” and “final_score” columns together
  - Show only students with “final_score” is greater than 70
  - Add a new column: “bonus_score” whose values are: final_score + 5
  - Add a new column: “passed” (True if final_score >= 60, False otherwise)

When you work with a DataFrame, always ask:

- *Does this operation change the original DataFrame?*
- *Does it return a new object (e.g., a Series, a DataFrame, or an Index)?*
- *Does it change the value of existing variables?*

Summary

- Jupyter Notebook Hands-on Practices
- DataFrame Operations
  - Re-index, rename, add, and drop
  - Select rows/columns, filter by conditions
  - Perform arithmetic
- Key Mindset
  - Practice is the best way to learn
  - Always ask: *Does this operation change the DataFrame or return something new?*
  - Don’t try to memorize everything – get comfortable through use
- Useful Resource
  - Pandas tutorial from W3Schools
- Individual Assignment 1
  - Due: **26 Sep 2025, 23:59**